

MH540	PPS Compound Glass Fiber Reinforced 聚苯硫醚复合材料 玻璃纤维增强
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Product Profile 产品概况

This product is a 40% glass fiber reinforced PPS compound with excellent strength, rigidity and processability.
本产品是40%的玻璃纤维增强的聚苯硫醚复合材料，具有优异的强度、刚性和成型加工性能。

Color 颜色	NC 本色, BK 黑色	Component 成分 ¹		PPS-GF40
Form 形式	Pellets 粒子	Processing Method 加工方式		Injection Molding 注塑成型
Status 状态	Commercially Available 已商用	Yellow Card 黄卡 ²		QMFZ2.E539835
Features 特征	High Temperature Resistance Inherent Flame Retardancy	耐高温 自阻燃	Excellent Chemical Resistance Good Electrical Properties	优异的耐化学品性 良好的电气性能
Application 用途	Automotive & Transportation Electrical & Electronics Office & Household Use	汽车与交通 电子电气 办公和家用	Energy & Electricity Industrial & Manufacturing Aerospace	能源与电力 工业和制造 航空和航天

Technical Data 技术参数

Physical Properties 物理性质	Value 数值	Unit 单位	Test Method 测试方法
Density 密度	1.66	g/cm ³	ISO 1183-1
Water Absorption, 23 °C /24hrs. 吸水率, 23 °C /24hrs.	0.02	%	ISO 62
Mold Shrinkage ³ 成型收缩率 ³	0.3/0.7	%	ISO 294-4
Mechanical Properties 机械性能	Value 数值	Unit 单位	Test Method 测试方法
Tensile Strength (5mm/min), 23 °C 拉伸强度, 23 °C	200	MPa	ISO 527-1/-2
Tensile Modulus (1mm/min), 23 °C 拉伸模量, 23 °C	15.5	GPa	ISO 527-1/-2
Tensile Strain at Break (5mm/min), 23 °C 拉伸断裂应变, 23 °C	2.0	%	ISO 527-1/-2
Flexural Strength (2mm/min), 23 °C 弯曲强度, 23 °C	285	MPa	ISO 178
Flexural Modulus (2mm/min), 23 °C 弯曲模量, 23 °C	15.0	GPa	ISO 178
Charpy Impact Strength, Notched, 23 °C 简支梁冲击强度, 缺口, 23 °C	11	kJ/m ²	ISO 179/1eA
Charpy Impact Strength, Unnotched, 23 °C 简支梁冲击强度, 无缺口, 23 °C	58	kJ/m ²	ISO 179/1eU
Charpy Impact Strength, Notched, -30 °C 简支梁冲击强度, 缺口, -30 °C	11	kJ/m ²	ISO 179/1eA
Charpy Impact Strength, Unnotched, -30 °C 简支梁冲击强度, 无缺口, -30 °C	55	kJ/m ²	ISO 179/1eU

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Rockwell Hardness, R Scale 洛氏硬度, R标尺	120	-	ISO 2039-2
Rockwell Hardness, M Scale 洛氏硬度, M标尺	100	-	ISO 2039-2
Thermal Properties 热性能	Value 数值	Unit 单位	Test Method 测试方法
Deflection Temp. Under Load (DTUL), 1.80MPa 载荷下热变形温度, 1.80MPa	265	℃	ISO 75-1/-2
Deflection Temp. Under Load (DTUL), 0.45MPa 载荷下热变形温度, 0.45MPa	>270	℃	ISO 75-1/-2
Melting Temperature, (10 °C/min) 熔融温度, (10 °C/min)	282	℃	ISO 11357-1/-3
Coef. of Linear Thermal Expansion. ³ , -50~50 °C 线性热膨胀系数 ³ , -50~50 °C	1.5/4.0	×10 ⁻⁵ /K	ISO 11359-1/-2
Coef. of Linear Thermal Expansion. ³ , 50~100 °C 线性热膨胀系数 ³ , 50~100 °C	1.5/6.0	×10 ⁻⁵ /K	ISO 11359-1/-2
Coef. of Linear Thermal Expansion ³ , 100~200 °C 线性热膨胀系数 ³ , 100~200 °C	1.5/8.5	×10 ⁻⁵ /K	ISO 11359-1/-2
Flammability / Thickness (mm) 燃烧性 / 厚度 (mm)	V-0/0.75	-	UL 94
Limiting oxygen index (LOI) 极限氧指数	47	%	ISO 4589
Electrical Properties 电性能	Value 数值	Unit 单位	Test Method 测试方法
Electric Strength 介电强度	28	kV/mm	IEC 60243-1
Relative Permittivity, 1MHz 相对介电常数, 1MHz	3.6	-	IEC 62631-2-1
Dissipation Factor, 1MHz 介电损耗因数, 1MHz	3.0	×10 ⁻³	IEC 62631-2-1
Relative Permittivity, 2.5GHz/5GHz 相对介电常数, 2.5GHz/5GHz	3.9/4.0	-	IEC 61189-2-721
Dissipation Factor, 2.5GHz/5GHz 介电损耗因数, 2.5GHz/5GHz	4.8/5.5	×10 ⁻³	IEC 61189-2-721
Comparative Tracking Index (CTI) 相对耐漏电起痕指数 (CTI)	175 (3)	V (PLC)	IEC 60112 (UL746 A)
Volume Resistivity 体积电阻率	10 ¹⁴	Ω·m	IEC 62631-3-1
Surface Resistivity 表面电阻率	10 ¹⁶	Ω	IEC 62631-3-2

Note 注释

1. In compliance with ISO 1043 符合ISO 1043标准
2. <https://iq.ulprospector.com/zh-cn/profile?e=7569313>
3. FD/TD, Flow Direction 流动方向 / Transverse Direction 垂直方向

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Typical Injection Molding Conditions 典型注塑成型工艺

Pre-drying 预干燥

120 ℃ / 4 h; or 130 ℃ / 3 h; or 140 ℃ / 2 h
120 ℃ / 4 小时; 或者 130 ℃ / 3 小时; 或者 140 ℃ / 2 小时

Mold Temp. 模具温度

130 ~ 150 ℃ (Recommended to use 140 ℃ and avoid the 80~120 ℃ range)
130 ~ 150 ℃ (建议使用140 ℃, 并避开80~120 ℃的范围)

Injection Conditions 注塑工艺

Barrel Profile 料筒温度	Resin (Melt) 树脂(熔体)	Nozzle 射嘴	Zone4 四区	Zone3 三区	Zone2 二区	Zone1 一区	Feed 喂料
Max (℃) 上限	340	330	340	330	320	300	100
Rec. (℃) 推荐值	320	315	320	315	310	290	-
Min (℃) 下限	300	300	300	300	300	280	-
Injection Pressure 注射压力	50 ~ 100 MPa		Pack Pressure 保压压力	30 ~ 80 MPa			
Injection Rate 注射速率	Medium to Fast 中高速		Pack Time 保压时间	3.0 ~ 10.0 s			
Injection Time 注射时间	0.5 ~ 2 s		Back Pressure 背压	0.3 ~ 3 MPa			
Screw Cushion 射胶终点	3 ~ 8 mm		Screw Speed 螺杆转速	50 ~ 100 RPM			

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